
AN OVERVIEW ON THE STRUCTURE OF THE WEIHRAUCH DEGREES

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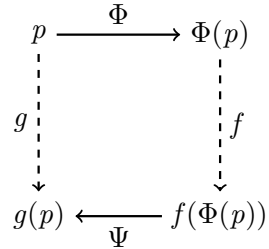
Dagstuhl meeting 25131

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REDUCIBILITIES ON COMPUTATIONAL PROBLEMS

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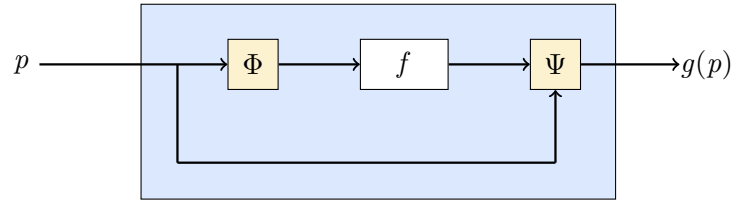
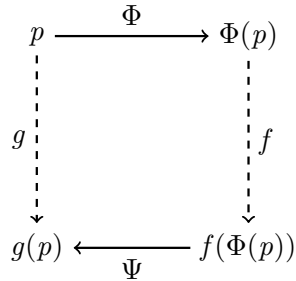
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I will refer to Φ and Ψ as the *forward* and *backward* functionals respectively.

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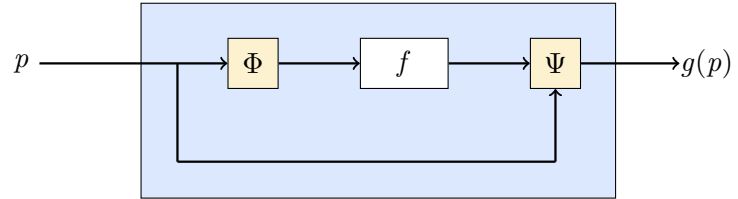
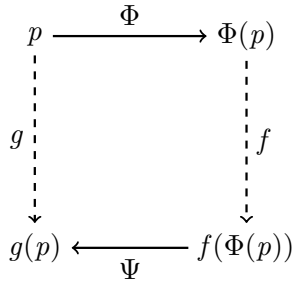
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According to the properties of Φ and Ψ we get different reductions.

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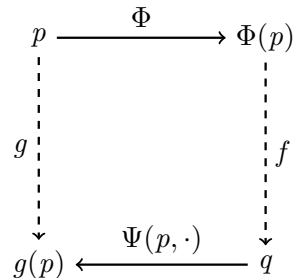
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$g \leq_W f \iff$ there are computable $\Phi, \Psi : \subseteq \mathbb{N}^{\mathbb{N}} \rightarrow \mathbb{N}^{\mathbb{N}}$ such that

- Given $p \in \text{dom}(g)$, $\Phi(p) \in \text{dom}(f)$
- Given $q \in f(\Phi(p))$, $\Psi(p, q) \in g(p)$



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Proposition (Higuchi, Pauly)

The Weihrauch degrees are neither a Heyting nor a Brouwer algebra, i.e.

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Every (non-zero) degree has $2^{\mathfrak{c}}$ -many degrees above and below it.

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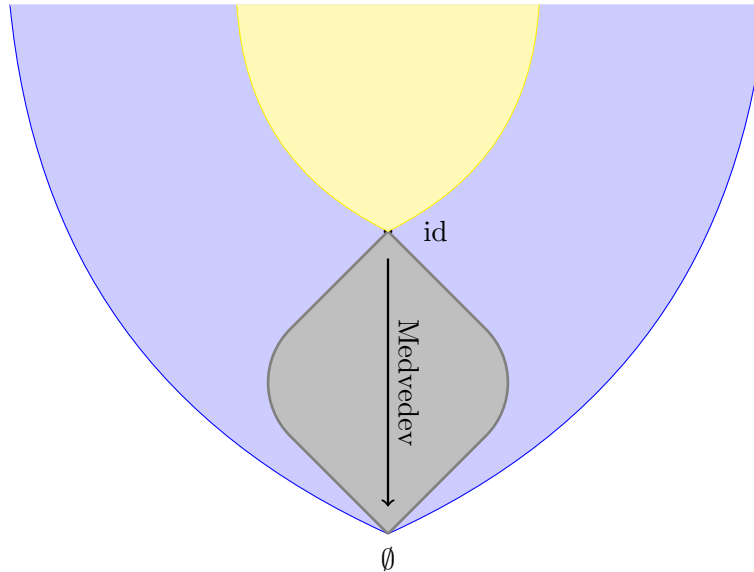
$$d_A : A \rightarrow \{0^{\mathbb{N}}\} := p \mapsto 0^{\mathbb{N}}$$

It follows that: $B \leq_M A$ iff $d_A \leq_W d_B$

This gives an embedding of the Medvedev degrees (upside down) in the lower cone of id .

The Medvedev degrees can be embedded also directly (i.e. without reversing the order), but the range of the embedding is harder to describe.

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Corollary (of Terwijn)

Under $\text{ZFC} + 2^{<\mathfrak{c}} = \mathfrak{c}$, there is a chain of size $2^{\mathfrak{c}}$ in $\mathcal{W}$.

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This is a special case of a more general characterization describing when a family of problems admit an upper bound.

Theorem (Lempp, Marcone, V.)

Let $\kappa > \omega$ be a regular cardinal and let $(f_\alpha)_{\alpha < \kappa}$ be a chain of multi-valued functions. TFAE:

1. $(f_\alpha)_{\alpha < \kappa}$ has an upper bound in $\mathcal{W}$;
2. there is a cofinal subsequence $(\bar{f}_\alpha)_{\alpha < \kappa}$ such that, for every $p \in \bigcup_{\alpha < \kappa} \text{dom}(\bar{f}_\alpha)$,

$$\bigcap \{\bar{f}_\alpha(p) : p \in \text{dom}(\bar{f}_\alpha)\} \neq \emptyset.$$

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The previous characterization does not give any example of sequences without upper bound.

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Let $\mathcal{P}$ be a poset. $Q \subseteq \mathcal{P}$ is **cofinal in $\mathcal{P}$** if every element of $\mathcal{P}$ is below some element of Q .

The **set-cofinality** of $\mathcal{P}$ is the smallest size of a cofinal **subset**.

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This is in stark contrast with the Medvedev degrees, where the set-cofinality is $\mathfrak{c}$ and the existence of cofinal chains is equivalent to CH.

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While large antichains can be found everywhere, extendible antichains are harder to characterize.

Theorem (Lempp, Marcone, V.)

Let $\{f_\alpha\}_{\alpha < \mathfrak{c}}$ be an antichain in $\mathcal{W}$ and let $D_\alpha := \text{dom}(f_\alpha)$. If the family

$$\mathcal{B} := \{D_\beta : (\forall x \in D_\beta)(f_\beta(x) \text{ has a computable element})\}$$

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Corollary (Lempp, Marcone, V.)

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Question: is there a maximal antichain in $\mathcal{W}$ of size $\mathfrak{c}$?

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There are no minimal degrees in the Weihrauch degrees.

In fact, every (non-trivial) initial segment of $\mathcal{W}$ has size $2^{\mathfrak{c}}$ (as that is the size of lower cones).

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Question: what are the possible order types of chains in $\mathcal{W}$ with no non-zero lower bound?

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In a poset, **a** is a **minimal cover** of **b** if $\{\mathbf{c} : \mathbf{b} < \mathbf{c} < \mathbf{a}\} = \emptyset$ (i.e. the interval $(\mathbf{a}, \mathbf{b})$ is empty).
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Empty intervals in the Medvedev degrees have been fully characterized:

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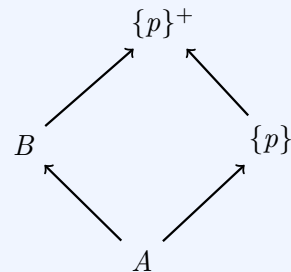
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Theorem (Dyment)

For every $A <_{\mathbf{M}} B$, B is a minimal cover of A iff

$$(\exists p \in A)[A \equiv_{\mathbf{M}} B \wedge \{p\} \text{ and } B \wedge \{p\}^+ \equiv_{\mathbf{M}} B],$$

where $P \wedge Q := (0) \wedge P \cup (1) \wedge Q$ is the meet in the Medvedev degrees.



EMPTY INTERVALS AND STRONG MINIMAL COVERS

Corollary

For every $p \in \mathbb{N}^{\mathbb{N}}$, $\text{id}_{\{p\}}$ is a strong minimal cover of $\text{id}_{\{p\}^+}$.

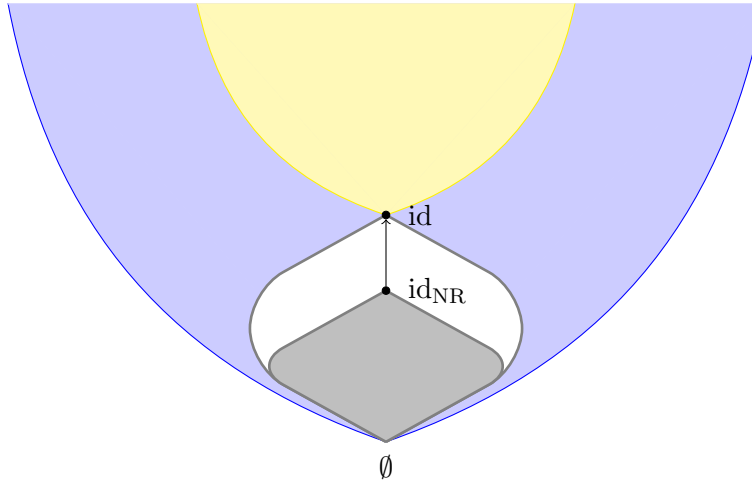
In particular, id is a SMC of id_{NR} , where $\text{NR} := \{q : q \not\leq_T 0\} \equiv_{\text{M}} \{0^{\mathbb{N}}\}^+$.

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STRONG MINIMAL COVERS IN WEIHRAUCH

Theorem (Lempp, Miller, Pauly, Soskova, V.)

The following are equivalent:

1. f is a strong minimal cover of h in the Weihrauch degrees.
2. $h \equiv_W \text{id}_{\{p\}^+}$ and $f \equiv_W \text{id}_{\{p\}}$ for some $p \in \mathbb{N}^{\mathbb{N}}$.

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The Weihrauch degree of id is the greatest degree that is a strong minimal cover. In particular, it is first-order definable in $(\mathcal{W}, \leq_W)$.

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Corollary (Lempp, Miller, Pauly, Soskova, V.)

The first-order theory of the Weihrauch degrees, the first-order theory of the Weihrauch degrees below id , and the third-order theory of true arithmetic are pairwise recursively isomorphic.

MINIMAL COVERS IN WEIHRAUCH

Theorem (Lempp, Miller, Pauly, Soskova, V.)

Let f and h be partial multi-valued functions on Baire space. The following are equivalent:

1. f is a minimal cover of h in the Weihrauch degrees.
2. $f \equiv_W h \sqcup \text{id}_{\{p\}}$ for some p with $\text{dom}(h) \not\leq_M \{p\}$ and $\text{dom}(h) \leq_M \{p\}^+$.

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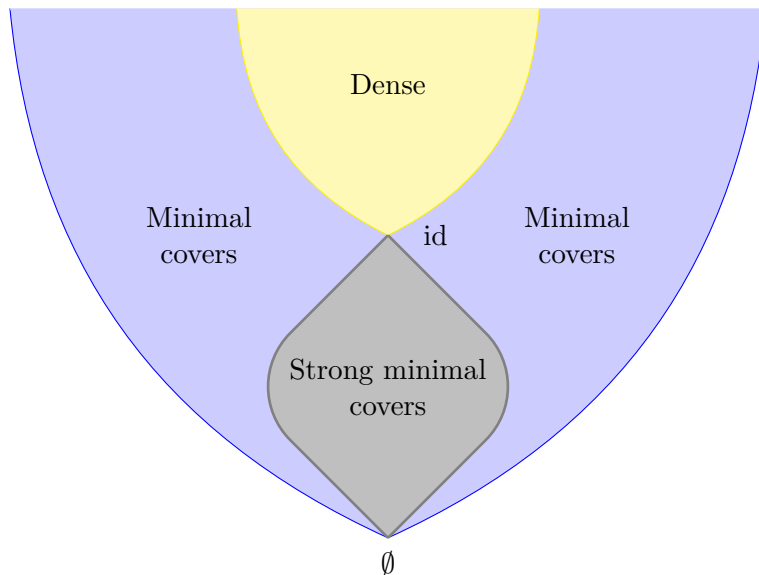
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Corollary (Lempp, Miller, Pauly, Soskova, V.)

The Weihrauch degrees above id are dense. In fact, id is the least degree whose upper Weihrauch cone is dense.

OVERVIEW



MINIMAL COVERS AND TURING-ANTICHAINS

Theorem (Lempp, Miller, Pauly, Soskova, V.)

For every family of pairwise Turing incomparable sets $\{p_\alpha\}_{\alpha < \kappa}$ with $\kappa < \mathfrak{c}$, there is a multi-valued function h whose minimal covers are exactly those of the form $h \sqcup \text{id}_{\{p_\alpha\}}$.

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Corollary (Lempp, Miller, Pauly, Soskova, V.)

For every cardinal $\kappa \leq \mathfrak{c}$, there is a problem h with exactly κ minimal covers.

EMBEDDINGS

The multi-valued functions in the cone above id are exactly the **pointed** problems, i.e. the problems that have at least a computable input (the vast majority of problems studied in the literature).

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The range of such endomorphism is a (proper) subset of the pointed, join-irreducible degrees.

Question: does every (pointed) degree contain an isomorphic copy of the Weihrauch degrees in its upper cone?

MINIMAL PAIRS

In a poset, **a, b** form a **minimal pair above c** if whenever **d < a** and **d < b** then **d ≤ c**.
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Question: is there a characterization of the minimal pairs above a given degree?

INTERVALS

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Every (non-empty) interval in $\mathcal{M}$ is either isomorphic to the Boolean algebra 2^n for some $n \geq 1$ or contains an antichain of size 2^c .

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Theorem (Lempp, Marcone, V.)

Every infinite interval in $\mathcal{W}$ is uncountable.

Question: do all infinite intervals in $\mathcal{W}$ have size 2^c ?

Question: is there an $\omega + \omega^*$ chain in $\mathcal{W}$ with no intermediate degree?

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